

Using Large Language Models to Generate Authentic Multi-agent Knowledge Work Datasets

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Abstract: Current publicly available knowledge work data collections lack diversity, extensive annotations, and contextual information about the users and their documents. These issues hinder objective and comparable data-driven evaluations and optimizations of knowledge work assistance systems. Due to the considerable resources needed to collect such data in real-life settings and the necessity of data censorship, collecting such a dataset appears nearly impossible. For this reason, we propose a configurable, multi-agent knowledge work dataset generator. This system simulates collaborative knowledge work among agents producing Large Language Model-generated documents and accompanying data traces. Additionally, the generator captures all background information, given in its configuration or created during the simulation process, in a knowledge graph. Finally, the resulting dataset can be utilized and shared without privacy or confidentiality concerns.

This paper introduces our approach’s design and vision and focuses on generating authentic knowledge work documents using Large Language Models. Our study involving human raters who assessed 53% of the generated and 74% of the real documents as realistic demonstrates the potential of our approach. Furthermore, we analyze the authenticity criteria mentioned in the participants’ comments and elaborate on potential improvements for identified common issues.

Keywords: Knowledge Work Dataset Generator, Large Language Models, Multi-agent Simulation

1 Introduction

Conducting data-driven evaluations of knowledge work support tools allows for deeper, objective insights into the tools’ weaknesses, strengths, and their reasons compared to user studies. Moreover, it enables meaningful comparisons that are reproducible from the input data perspective.

However, currently, there are no comprehensive knowledge work datasets available that fully satisfy desirable properties. Gonçalves [Go11] addressed this problem more than ten years ago and identified the incompleteness, lack of background information about the users and their documents, limited generalization, and the lack of ground truth information to base evaluations on as key issues of existing datasets. To address these issues, he envisioned

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a large-scale data collection from a representative set of users who must annotate their data extensively. However, he also mentioned the difficulty of realizing this vision.

A recent publication of Bakhshizadeh et al. [Ba24] shows that these problems persist. They introduced a data collection named RLKWiC addressing the issues of missing background information by having eight participants track and annotate their daily work for two months and categorizing it into self-defined and annotated contexts. While the RLKWiC dataset represents a step towards satisfying Gonçalves’ [Go11] vision, it is still specific to context-related tasks and affected by data incompleteness due to censoring required to preserve privacy and confidentiality.

Given the challenges of collecting comprehensive, heavily annotated datasets and their persistent issues with adaptability, generalization, scalability, and data incompleteness due to censoring, we propose KnoWoGen, a knowledge work dataset generator. KnoWoGen simulates multiple knowledge workers solving tasks, creating documents, and collaborating. While a Multi-agent System handles the task distribution and scheduling, documents are generated by prompting Large Language Models. Moreover, our solution enables engineers of knowledge work assistance tools to configure and create a dataset fitting their evaluation needs. Additionally, resulting datasets can be publicly shared for comparison. All background information given in the configuration or introduced during the simulation is stored in a knowledge graph to preserve all contextually relevant information that can be later used as input, ground truth information, or for analysis purposes when evaluating assistance tools on the generated dataset.

This paper is organized as follows: In Section 2, we discuss related work. In Section 3, we present the design of our KnoWoGen approach. Section 4 examines the gap between generated and real knowledge work documents and provides an in-depth analysis of decisive criteria. Section 5 discusses the experiment results and derived improvement potentials. Finally, Section 6 summarizes our findings and outlines potential future work.

2 Related Work

There are multiple generators related to our approach. Guo et al. [GPH05] introduced the knowledge graph generator LUBM, a customizable, scalable solution for triple store benchmarking. Other generators target specific document types related to knowledge work, such as spreadsheets [SJD21] or emails [Ba18]. The approach of Schulze et al. [Sc21] is also similar to the KnoWoGen. It simulates business processes resulting in accounting documents, such as invoices, and a knowledge graph modeling the processes and associations with documents. This approach and the previously introduced document generators have in common that they synthesize documents by modifying examples or working with blueprints and template patterns. Consequently, the diversity of generated documents is limited. Recently, with the emergence of Large Language Models (LLMs), new document generators

evolved based on them (see, e.g., [Ka23; Mo24]) which, in contrast to earlier solutions, can generate more diverse documents without requiring explicit templates or examples.

Moreover, several simulators based on LLM-operated Multi-agent Systems emerged. Wang et al. [Wa24] and Guo et al. [Gu24] provide an overview of these approaches. In contrast to typical Multi-agent Systems, the agents' subsequent actions are not decided randomly or by following a predefined concrete strategy with a set of rules. Instead, the LLM decides and executes the following actions.

Our approach also utilizes LLMs to generate documents reflecting the task-solving actions of one or multiple knowledge workers. However, the LLM is not responsible for deciding which successive actions an agent will perform. Nevertheless, the generated documents influence the outcome of subsequent actions in which, for instance, a document is synthesized based on the content of a previous one. Additionally, our tool generates a corresponding knowledge graph that, amongst others, also puts the generated documents in relation.

Since we wanted to evaluate how authentic generated knowledge work documents are perceived by humans and which differences to real documents exist, we conducted an experiment inspired by the Turing test [Tu50]. Hence, we follow the examples of, for instance, Jannai et al. [Ja23] who examined whether humans can distinguish human- and AI chat partners, Clark et al. [Cl21] conducting a similar test on recipes, stories, and news articles and Jakesch et al. [JHN23] analyzing generated and real self-introductions.

3 Methodology

This section introduces the design of our knowledge work dataset generator KnoWoGen. First, we provide an overview and then explain the simulation setup and process in more detail.

Overview: KnoWoGen is a generator framework that synthesizes knowledge work datasets by simulating the collaborative knowledge work of multiple agents. A knowledge work assistance tool engineer needs to configure the KnoWoGen accordingly to create a suitable dataset to evaluate or optimize their tool. Based on these settings, KnoWoGen sets up the simulation environment with agents and tasks. Subsequently, in each simulation round, agents get tasks assigned and conduct them, resulting in either documents or data traces. The KnoWoGen stores all information about the simulation setting and completed tasks in a knowledge graph.

Simulating Knowledge Workers and Their Environment: The configuration sets up the simulation environment. It encompasses the specification of agents, companies, departments, and the domain. In the current implementation, agents are defined by their names and job